

Woojae Shin

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Born June 26, 1998

Updated October 2026

Ph.D. Student in Mechanical Engineering

Korea Advanced Institute of Science and Technology (KAIST)

Profile

Ph.D. student at KAIST researching reinforcement learning for autonomous robotic systems, particularly unmanned aerial vehicles. Research interests also include robotic perception, state estimation, and multi-robot systems.

Education

2025 - Present	Ph.D. Student in Mechanical Engineering Korea Advanced Institute of Science and Technology (KAIST) Laboratory: Field AI and Robotics (FAIR) Laboratory
2025	M.S. in Mechanical Engineering Ulsan National Institute of Science and Technology (UNIST) Laboratory: Autonomous Systems Lab (ASL)
2023	B.S. in Mechanical Engineering Ulsan National Institute of Science and Technology (UNIST)

Awards and competitions

2026	AI Grand Prix — Team KAIST-FAIR [Virtual result] [Physical recap] 1st place, Virtual Qualifier 2; 2nd place, Physical Qualifier Autonomous drone racing competition Qualified for the Season 1 Championship, scheduled for December 11, 2026; championship not yet contested.
2024	ICUAS 2024 UAV Competition — Team UNIST ASL [Official result] Winner International Conference on Unmanned Aircraft Systems, Chania, Greece Crop-yield estimation in indoor and outdoor agricultural scenarios.

Research interests

Reinforcement learning / Autonomous UAVs / Robotic perception / State estimation / Multi-robot systems

Publications and presentations

International journal articles (5)

2026	Vision-Based Autonomous Drone Landing on Moving Platforms With Uncertain Motion via Deep Reinforcement Learning [DOI] [Project] Woojae Shin, Minwoo Kim, Taewook Park, Geunsik Bae, Seunghwan Kim, and Hyondong Oh IEEE Robotics and Automation Letters , 11(5), 5542-5549
2026	Real-Time Communication Relay Planning With a Low-Complexity Network Quality Prediction Model in Dynamic Indoor Missions [DOI] Jaemin Seo, Jongyun Kim, Seunghwan Kim, Changseung Kim, Woojae Shin, and Hyondong Oh IEEE Robotics and Automation Letters , 11(1), 322-329

- 2026 **PACMAN: Rapid Identification of Keypoint Patch-Based Fiducial Marker in Occluded Environments** [\[DOI\]](#)
Taewook Park, Geunsik Bae, **Woojae Shin**, Meraj Mammadov, Jaemin Seo, Heejung Shin, and Hyondong Oh
Image and Vision Computing, 165, 105821
- 2025 **EKF-Based Radar-Inertial Odometry With Online Temporal Calibration** [\[DOI\]](#)
Changseung Kim, Geunsik Bae, **Woojae Shin**, Sen Wang, and Hyondong Oh
IEEE Robotics and Automation Letters, 10(7), 7230-7237
- 2024 **Autonomous Landing on a Moving Platform Using Vision-Based Deep Reinforcement Learning** [\[DOI\]](#)
Pawel Ladosz, Meraj Mammadov, Heejung Shin, **Woojae Shin**, and Hyondong Oh
IEEE Robotics and Automation Letters, 9(5), 4575-4582

Domestic journal article (1)

- 2024 **Collective Navigation Through a Narrow Gap for a Swarm of UAVs Using Curriculum-Based Deep Reinforcement Learning** [\[DOI\]](#)
Myong-Yol Choi, **Woojae Shin**, Minwoo Kim, Hwi-Sung Park, Youngbin You, Min Lee, and Hyondong Oh
The Journal of Korea Robotics Society, 19(1), 117-129

Peer-reviewed conference papers (3)

- 2026 **Learning Point Cloud Geometry as a Statistical Manifold: Theory and Practice** [\[DOI\]](#)
Jinwoo Lee, Jiwoo Kim, **Woojae Shin**, Giseop Kim, and Hyondong Oh
Proceedings of Robotics: Science and Systems XXII, Sydney, Australia
- 2025 **RAPID: Robust and Agile Planner Using Inverse Reinforcement Learning for Vision-Based Drone Navigation** [\[DOI\]](#)
Minwoo Kim, Geunsik Bae, Jinwoo Lee, **Woojae Shin**, Changseung Kim, Myong-Yol Choi, Heejung Shin, and Hyondong Oh
Proceedings of Robotics: Science and Systems XXI, Los Angeles, CA, USA
- 2025 **Doppler Correspondence: Non-Iterative Scan Matching With Doppler Velocity-Based Correspondence** [\[DOI\]](#)
Jiwoo Kim, Geunsik Bae, Changseung Kim, Jinwoo Lee, **Woojae Shin**, and Hyondong Oh
Proceedings of Robotics: Science and Systems XXI, Los Angeles, CA, USA

Workshops and conference presentations (5)

- 2026 **Self-Supervised Geometry Reasoning for LiDAR Simultaneous Localization and Mapping** [\[Program\]](#)
Jiwoo Kim, Jinwoo Lee, **Woojae Shin**, Giseop Kim, and Hyondong Oh
From Perception to Action Workshop at RSS 2026, Sydney, Australia (non-archival)
- 2025 **Vision-Based Autonomous Drone Landing on Fast-Moving Platforms Via Deep Reinforcement Learning** [\[Program\]](#)
Woojae Shin, Minwoo Kim, Taewook Park, Geunsik Bae, Seunghwan Kim, and Hyondong Oh
13th International Conference on Robot Intelligence Technology and Applications, London, UK (presentation only)
- 2024 **ReDepth: Rectified Activations for Robust Depth Estimation** [\[Program\]](#)
Jiwoo Kim, **Woojae Shin**, Heejung Shin, Minwoo Kim, Jinwoo Lee, and Hyondong Oh
12th International Conference on Robot Intelligence Technology and Applications, Ulsan, Korea (poster)

- 2024 **RAPID: Robust and Agile Planner Using Inverse Reinforcement Learning for Drones**
Minwoo Kim, Geunsik Bae, Jinwoo Lee, **Woojae Shin**, Heejung Shin, Myong-Yol Choi, and Hyondong Oh
14th Workshop on Planning, Perception and Navigation for Intelligent Vehicles at IROS 2024, Abu Dhabi, UAE
- 2024 **Multi-State Constraint Radar-Inertial Odometry [Record]**
Changseung Kim, Geunsik Bae, **Woojae Shin**, and Hyondong Oh
IEEE International Conference on Robotics and Automation, Yokohama, Japan (poster)

Domestic conferences (6)

- 2026 **Design and Implementation of Mission Space Modeling System for Autonomous Cooperation of Multi-Domain Unmanned Vehicles [Record]**
Y. Jeong, H. Yoo, **Woojae Shin**, and Hyondong Oh
2026 Spring Conference of the Korean Society for Aeronautical and Space Sciences, 1689-1690
- 2026 **Real-Time Communication Relay Planning With a Low-Complexity Network Quality Prediction Model in Dynamic Indoor Missions [Program]**
Jaemin Seo, Jongyun Kim, Seunghwan Kim, Changseung Kim, **Woojae Shin**, and Hyondong Oh
21st Korea Robotics Society Annual Conference, Pyeongchang, Korea (presentation)
- 2025 **Efficient Scan Matching Leveraging Doppler Velocity [Record]**
Jiwoo Kim, Jinwoo Lee, Geunsik Bae, Changseung Kim, **Woojae Shin**, and Hyondong Oh
40th ICROS Annual Conference, 34-35
- 2025 **Vision-Based Drone Navigation Using Inverse Reinforcement Learning [Record]**
Minwoo Kim, Geunsik Bae, Jinwoo Lee, **Woojae Shin**, Changseung Kim, Myong-Yol Choi, Heejung Shin, and Hyondong Oh
40th ICROS Annual Conference, 153-154
- 2025 **Vision-Based Autonomous Landing on a Fast-Moving Platform Using Deep Reinforcement Learning**
Woojae Shin, Minwoo Kim, Taewook Park, and Hyondong Oh
20th Korea Robotics Society Annual Conference, Pyeongchang, Korea (presentation)
- 2024 **Real-Time Fruit Tracking Framework for Autonomous Drone-Based Crop Monitoring Systems [Record]**
Geunsik Bae, Jaemin Seo, Taewook Park, **Woojae Shin**, Jinwoo Lee, Hankyoul Ko, Minjae Jung, Hongro Jang, Myeonggeun Gu, and Hyondong Oh
2024 Autumn Conference of the Korean Society for Aeronautical and Space Sciences, 88

Thesis (1)

- 2025 **Vision-Based Autonomous Drone Landing on Highly Dynamic Moving Platforms via Deep Reinforcement Learning**
Woojae Shin
M.S. thesis, Ulsan National Institute of Science and Technology (UNIST), August 2025